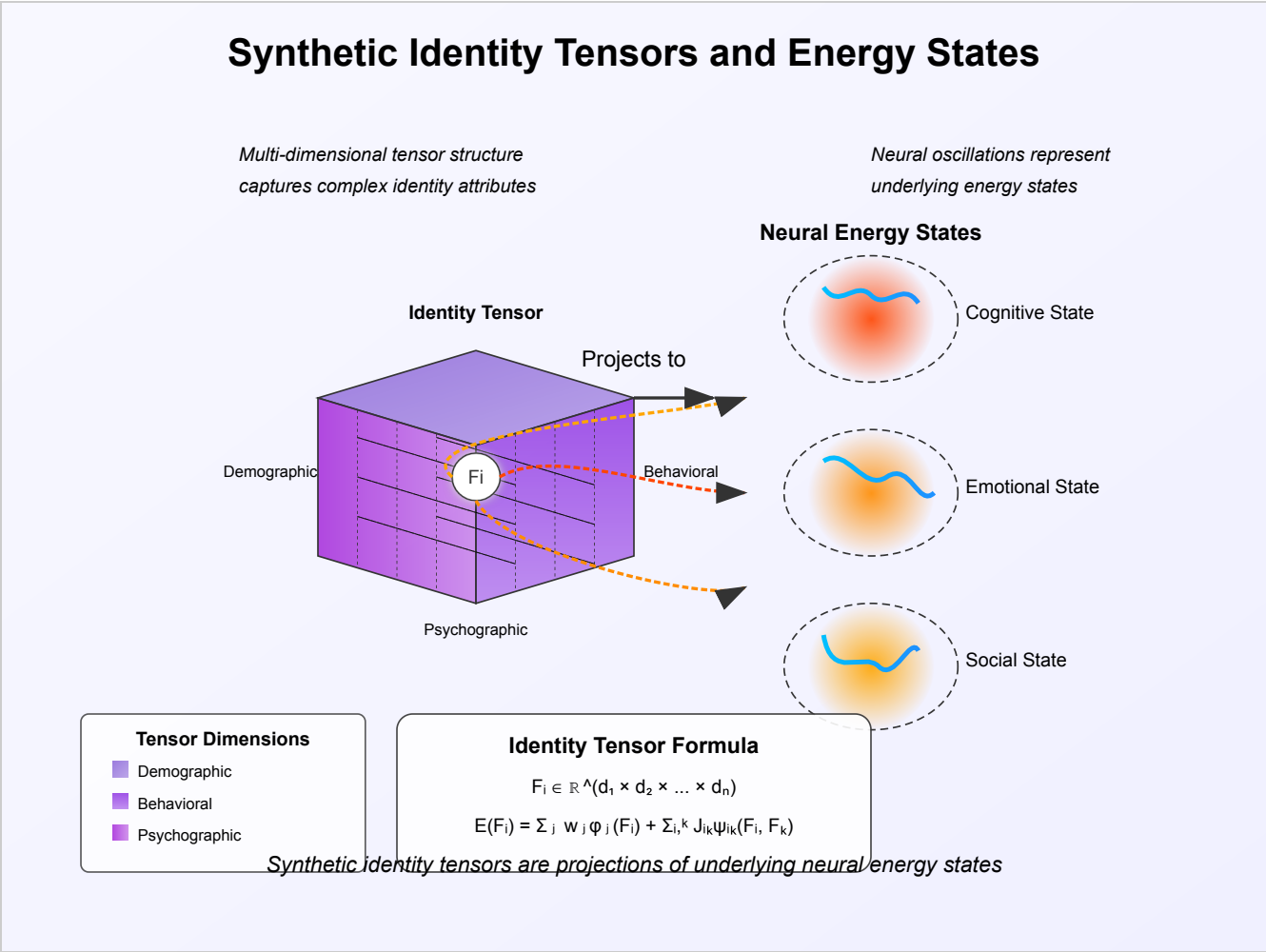

Identity in the proposed framework is represented not as a vector or matrix but as a multi-dimensional tensor that captures demographic, psychographic, behavioral, and relational attributes in a unified structure. These tensors exist within a high-dimensional feature space F_i where each synthetic identity (i) occupies a distinct region defined by both static and dynamic attributes.

These tensors enable the representation of complex, often contradictory aspects of identity that cannot be adequately captured in simpler models. By moving beyond token-based representations, these tensors can incorporate:

- Non-linguistic behavioral patterns
- Unconscious psychological drivers
- Cultural and contextual influences
- Temporal evolution of identity

1.2.2 Energy-Based Dynamics